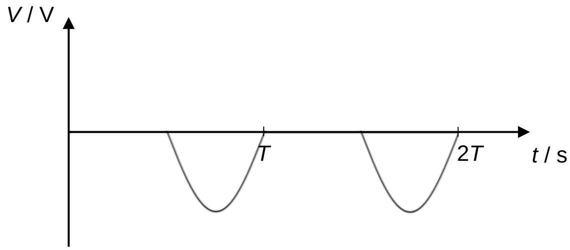
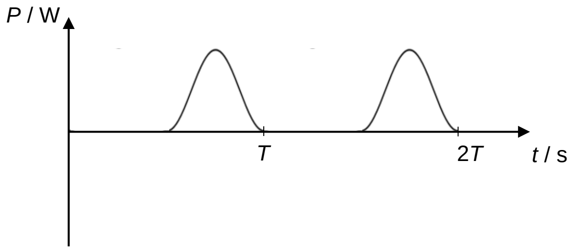
7	**(a)**	**(i)**		A1
		(ii)		A1

		(iii)	$P_{\max} = \dfrac{V_{\max}{}^2}{R} = \dfrac{5.0^2}{2.0} = 12.5\ \text{W}$ $P_{average} = \dfrac{P_{\max}}{4} = \dfrac{12.5}{4} = 3.1\ \text{W}$	M1 A1
	(b)	**(i)**	When an alternating current source is connected to the primary coil, there would be a changing magnetic flux produced. The iron-core strengthens and links the flux through the secondary coil. In the secondary coil, since the magnetic flux through it changes all the time, there would be an e.m.f. induced (according to Faraday's Law).	B1 B1
		(ii)	emf induced in secondary coil = 230 / 20 = 11.5 V current in secondary coil = V / R = 11.5 / 7.0 = 1.6429 A current in primary coil = 1.6429 / 20 = 0.082 A	 M1 A1

8	**(a)**		The cut-off wavelength corresponds to the most energetic photon	A1